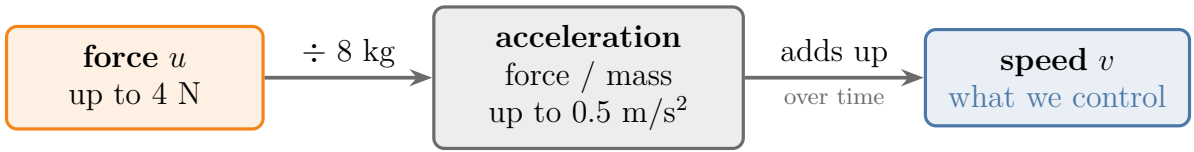
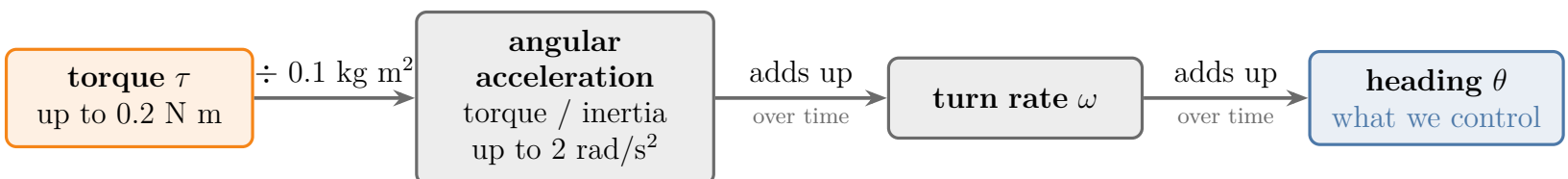
driving: one step from the motors to the speed



turning: two steps from the motors to the heading



at $\theta = 45°$ the torque can be 0,
yet ω is still large: the
robot keeps turning